Cisco Packet Tracer — Lab Report

3-Tier Hierarchical Enterprise Network

Status: COMPLETED

1. Network Overview

The network is structured in three hierarchical layers (Core, Distribution, Access) following the Cisco 3-tier model. OSPF area 0 is used as the routing protocol across all routers. Point-to-point links use /30 subnets, while endpoint VLANs use /24 networks. The command 'no ip domain-lookup' is configured on all routers to prevent unwanted DNS lookups.

Layer	Devices	PT Model	Role
Core	C1, C2	Router 2911	WAN connectivity + routing between distribution routers
Distribution	R1, R2, R3	Router 2911	Inter-VLAN routing + access layer connectivity
Access L3	S1, S2, S3	Switch 3560	Multilayer switches with SVI and OSPF
Access L2	A1-A6	Switch 2960	Access switches with VLAN segmentation

2. VLAN Design

VLAN	Department	Subnet	Gateway	Switch L3
VLAN 10	IT	192.168.10.0/24	192.168.10.1	S1
VLAN 11	IT Failover (DHCP)	192.168.11.0/24	192.168.11.1	S1
VLAN 20	HR	192.168.20.0/24	192.168.20.1	S2
VLAN 30	Administration	192.168.30.0/24	192.168.30.1	S3

3. Core Layer — C1 and C2

3.1 Topology

C1 and C2 are connected via serial DCE cable to the Cloud-PT (WAN), both sharing the same 203.0.113.0/30 subnet. The Cloud-PT bridges the two serial links internally via Frame Relay. A direct GigabitEthernet link between C1 and C2 provides redundancy. Both routers have clock rate 64000 on their serial interfaces.

Link	Subnet	C1 IP	C2 IP

C1 — WAN (Se0/2/0)	203.0.113.0/30	203.0.113.1	—
C2 — WAN (Se0/2/0)	203.0.113.0/30	—	203.0.113.2
C1 — C2 (Gig0/1)	10.0.0.0/30	10.0.0.1	10.0.0.2
C1 — R1 (Gig0/0)	10.1.1.0/30	10.1.1.1	—
C1 — R2 (Gig0/2)	10.1.2.0/30	10.1.2.1	—
C2 — R2 (Gig0/0)	10.1.3.0/30	—	10.1.3.1
C2 — R3 (Gig0/1)	10.1.4.0/30	—	10.1.4.1

3.2 C1 Configuration

```
hostname C1
no ip domain-lookup
interface Se0/2/0
 ip address 203.0.113.1 255.255.255.252
 clock rate 64000
 ip nat outside
 no shutdown
interface Gig0/0
 ip address 10.1.1.1 255.255.255.252
 ip nat inside
 no shutdown
interface Gig0/1
 ip address 10.0.0.1 255.255.255.252
 ip nat inside
 no shutdown
interface Gig0/2
 ip address 10.1.2.1 255.255.255.252
 ip nat inside
 no shutdown
router ospf 1
 network 10.0.0.0 0.0.0.3 area 0
 network 10.1.1.0 0.0.0.3 area 0
 network 10.1.2.0 0.0.0.3 area 0
 default-information originate
ip route 0.0.0.0 0.0.0.0 203.0.113.2
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.11.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
access-list 1 permit 192.168.30.0 0.0.0.255
ip nat inside source list 1 interface Se0/2/0 overload
```

3.3 C2 Configuration

```
hostname C2
no ip domain-lookup
interface Se0/2/0
 ip address 203.0.113.2 255.255.255.252
 clock rate 64000
 ip nat outside
 no shutdown
interface Gig0/0
 ip address 10.1.3.2 255.255.255.252
 ip nat inside
 no shutdown
interface Gig0/1
 ip address 10.1.4.2 255.255.255.252
 ip nat inside
 no shutdown
interface Gig0/2
 ip address 10.0.0.2 255.255.255.252
 ip nat inside
 no shutdown
router ospf 1
 network 10.0.0.0 0.0.0.3 area 0
 network 10.1.3.0 0.0.0.3 area 0
 network 10.1.4.0 0.0.0.3 area 0
 default-information originate
ip route 0.0.0.0 0.0.0.0 203.0.113.1
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.11.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
access-list 1 permit 192.168.30.0 0.0.0.255
ip nat inside source list 1 interface Se0/2/0 overload
```

4. Distribution Layer — R1, R2, R3

Link	Subnet	Router IP	Core IP
R1 Gig0/0 — C1	10.1.1.0/30	10.1.1.2	10.1.1.1 (C1)
R1 Gig0/1 — S1	10.2.1.0/30	10.2.1.1	10.2.1.2 (S1)
R2 Gig0/0 — C1	10.1.2.0/30	10.1.2.2	10.1.2.1 (C1)
R2 Gig0/1 — C2	10.1.3.0/30	10.1.3.2	10.1.3.1 (C2)
R2 Gig0/2 — S2	10.2.2.0/30	10.2.2.1	10.2.2.2 (S2)
R3 Gig0/0 — C2	10.1.4.0/30	10.1.4.2	10.1.4.1 (C2)
R3 Gig0/1 — S3	10.2.3.0/30	10.2.3.1	10.2.3.2 (S3)

4.1 R1 Configuration

```
hostname R1
no ip domain-lookup
interface Gig0/0
 ip address 10.1.1.2 255.255.255.252
 no shutdown
interface Gig0/1
 ip address 10.2.1.1 255.255.255.252
 no shutdown
router ospf 1
 network 10.1.1.0 0.0.0.3 area 0
 network 10.2.1.0 0.0.0.3 area 0
```

4.2 R2 Configuration

```
hostname R2
no ip domain-lookup
interface Gig0/0
 ip address 10.1.2.2 255.255.255.252
 no shutdown
interface Gig0/1
 ip address 10.1.3.2 255.255.255.252
 no shutdown
interface Gig0/2
 ip address 10.2.2.1 255.255.255.252
 no shutdown
router ospf 1
 network 10.1.2.0 0.0.0.3 area 0
 network 10.1.3.0 0.0.0.3 area 0
```

```
 network 10.2.2.0 0.0.0.3 area 0
```

4.3 R3 Configuration

```
hostname R3
no ip domain-lookup
interface Gig0/0
 ip address 10.1.4.2 255.255.255.252
 no shutdown
interface Gig0/1
 ip address 10.2.3.1 255.255.255.252
 no shutdown
router ospf 1
 network 10.1.4.0 0.0.0.3 area 0
 network 10.2.3.0 0.0.0.3 area 0
```

5. Access Layer L3 — S1, S2, S3 (Switch 3560)

The 3560 multilayer switches act as VLAN gateways via SVIs and participate in OSPF to propagate VLAN networks upstream.

Switch	Uplink to Router	VLANs	VLAN Gateways
S1	Gig0/1 -> R1 (10.2.1.2/30)	VLAN 10, VLAN 11	192.168.10.1 / 192.168.11.1
S2	Gig0/1 -> R2 (10.2.2.2/30)	VLAN 20	192.168.20.1
S3	Gig0/1 -> R3 (10.2.3.2/30)	VLAN 30	192.168.30.1

5.1 S1 Configuration

```
hostname S1
ip routing
interface Gig0/1
 no switchport
 ip address 10.2.1.2 255.255.255.252
 no shutdown
interface vlan 10
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 192.168.10.2
 ip helper-address 192.168.11.2
 no shutdown
interface vlan 11
 ip address 192.168.11.1 255.255.255.0
 no shutdown
interface Fa0/1
 switchport mode access
 switchport access vlan 10
 no shutdown
interface Fa0/2
 switchport mode access
 switchport access vlan 11
 no shutdown
router ospf 1
 network 10.2.1.0 0.0.0.3 area 0
 network 192.168.10.0 0.0.0.255 area 0
 network 192.168.11.0 0.0.0.255 area 0
```

5.2 S2 Configuration

```
hostname S2
ip routing
interface Gig0/1
```

```
 no switchport
 ip address 10.2.2.2 255.255.255.252
 no shutdown
interface vlan 20
 ip address 192.168.20.1 255.255.255.0
 ip helper-address 192.168.10.2
 ip helper-address 192.168.11.2
 no shutdown
interface Fa0/1
 switchport mode access
 switchport access vlan 20
 no shutdown
router ospf 1
 network 10.2.2.0 0.0.0.3 area 0
 network 192.168.20.0 0.0.0.255 area 0
```

5.3 S3 Configuration

```
hostname S3
ip routing
interface Gig0/1
 no switchport
 ip address 10.2.3.2 255.255.255.252
 no shutdown
interface vlan 30
 ip address 192.168.30.1 255.255.255.0
 ip helper-address 192.168.10.2
 ip helper-address 192.168.11.2
 no shutdown
interface Fa0/1
 switchport mode access
 switchport access vlan 30
 no shutdown
router ospf 1
 network 10.2.3.0 0.0.0.3 area 0
 network 192.168.30.0 0.0.0.255 area 0
```

6. Access Layer L2 — A1-A6 (Switch 2960)

The 2960 switches are pure Layer 2 devices. Each port is configured in access mode on the correct VLAN. No routing or OSPF is required.

Switch	VLAN	Department	Connected Endpoints
A1	10	IT	Printer, PC1, PC0
A2	10	IT	DNS Server, FTP/NAS Server, Primary DHCP Server
A3	11	IT Failover	Secondary DHCP Server
A4	20	HR	PC2, PC3
A5	30	Administration	Laptop0, PC4, Web Server
A6	30	Administration	PC5, PC6

6.1 A1 Configuration (example)

```
hostname A1
interface Fa0/1
 switchport mode access
 switchport access vlan 10
 no shutdown
interface Fa0/2
 switchport mode access
 switchport access vlan 10
 no shutdown
interface Fa0/3
 switchport mode access
 switchport access vlan 10
 no shutdown
```

Repeat for A2-A6 changing hostname and VLAN according to the table above.

7. Endpoint IP Addresses

Device	VLAN	IP Address	Subnet Mask	Gateway	DNS	Type
Primary DHCP Server	10	192.168.10.2	255.255.255.0	192.168.10.1	192.168.10.3	Static
DNS Server	10	192.168.10.3	255.255.255.0	192.168.10.1	192.168.10.3	Static
FTP/NAS Server	10	192.168.10.4	255.255.255.0	192.168.10.1	192.168.10.3	Static
Secondary DHCP Server	11	192.168.11.2	255.255.255.0	192.168.11.1	192.168.10.3	Static
Web Server (Intranet)	30	192.168.30.2	255.255.255.0	192.168.30.1	192.168.10.3	Static
PC0, PC1, Printer	10	DHCP (.10+)	255.255.255.0	192.168.10.1	192.168.10.3	Dynamic
PC2, PC3	20	DHCP (.10+)	255.255.255.0	192.168.20.1	192.168.10.3	Dynamic
PC4, PC5, PC6, Laptop0	30	DHCP (.10+)	255.255.255.0	192.168.30.1	192.168.10.3	Dynamic

8. DHCP Configuration

8.1 Primary DHCP Server (192.168.10.2)

Go to: Server > Services > DHCP. Modify the default 'serverPool' setting Start IP to .10. Add the following pools:

Pool	Default Gateway	DNS Server	Start IP	Subnet Mask
VLAN10	192.168.10.1	192.168.10.3	192.168.10.10	255.255.255.0
VLAN20	192.168.20.1	192.168.10.3	192.168.20.10	255.255.255.0
VLAN30	192.168.30.1	192.168.10.3	192.168.30.10	255.255.255.0

8.2 Secondary DHCP Server (192.168.11.2)

Configure the same pools as the primary server. Failover is simulated via dual ip helper-address on S1, S2, S3.

8.3 DHCP Relay

Configured on the SVIs of S1, S2, S3:

```
ip helper-address 192.168.10.2   ! Primary DHCP server
ip helper-address 192.168.11.2   ! Secondary DHCP server (failover)
```

9. Services Configuration

9.1 DNS Server (192.168.10.3)

Go to: Server > Services > DNS > On. Add the following A records:

Name	Type	IP Address
intranet.azienda.local	A Record	192.168.30.2
ftp.azienda.local	A Record	192.168.10.4
dhcp.azienda.local	A Record	192.168.10.2

Verify: nslookup intranet.azienda.local from any PC.

9.2 FTP/NAS Server (192.168.10.4)

Go to: Server > Services > FTP > On. Configure the following users:

Username	Password	Permissions
admin	admin	Read, Write, Delete, Rename, List
user	user	Read, List

Verify from Command Prompt: ftp 192.168.10.4

9.3 Web Server / Intranet (192.168.30.2)

Go to: Server > Services > HTTP > On. Enable HTTP and HTTPS. Customize index.html with intranet content.

Verify from Web Browser: http://192.168.30.2 or http://intranet.azienda.local

10. NAT Configuration

NAT overload (PAT) is configured on both C1 and C2. Internal networks are translated to the public IP address of the WAN serial interface. C1 and C2 share the 203.0.113.0/30 subnet for the WAN link.

Interface	NAT Role	Router
Se0/2/0	ip nat outside	C1 and C2
Gig0/0, Gig0/1, Gig0/2	ip nat inside	C1 and C2

```
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.11.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
```

```
access-list 1 permit 192.168.30.0 0.0.0.255
ip nat inside source list 1 interface Se0/2/0 overload
```

Verify: show ip nat translations | show ip nat statistics

11. Inter-VLAN ACLs

11.1 Security Policy

Source	Destination	Allowed	Reason
VLAN 10 (IT)	All VLANs	YES	IT has full network access
VLAN 20 (HR)	DNS 192.168.10.3	YES	Required for name resolution
VLAN 20 (HR)	DHCP 192.168.10.2	YES	Required to obtain IP address
VLAN 20 (HR)	Web Server 192.168.30.2	YES	HR accesses the intranet
VLAN 20 (HR)	VLAN 10 (IT)	YES	Return traffic to IT
VLAN 20 (HR)	FTP 192.168.10.4	NO	NAS managed by IT only
VLAN 20 (HR)	VLAN 30 (Admin)	NO	Administrative data is confidential
VLAN 30 (Admin)	DNS 192.168.10.3	YES	Required for name resolution
VLAN 30 (Admin)	DHCP 192.168.10.2	YES	Required to obtain IP address
VLAN 30 (Admin)	Web Server 192.168.30.2	YES	Admin manages the intranet
VLAN 30 (Admin)	VLAN 10 (IT)	YES	Return traffic to IT
VLAN 30 (Admin)	FTP 192.168.10.4	NO	NAS managed by IT only
VLAN 30 (Admin)	VLAN 20 (HR)	NO	HR data is confidential

11.2 ACL_HR — applied on S2 (VLAN 20)

```
ip access-list extended ACL_HR
 10 permit ip 192.168.20.0 0.0.0.255 host 192.168.10.3
 20 permit ip 192.168.20.0 0.0.0.255 host 192.168.10.2
 30 permit ip 192.168.20.0 0.0.0.255 host 192.168.30.2
 35 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
 40 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
 50 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
 60 permit ip any any
interface vlan 20
 ip access-group ACL_HR in
```

11.3 ACL_ADMINISTRATION — applied on S3 (VLAN 30)

```
ip access-list extended ACL_ADMINISTRATION
 10 permit ip 192.168.30.0 0.0.0.255 host 192.168.10.3
 20 permit ip 192.168.30.0 0.0.0.255 host 192.168.10.2
 30 permit ip 192.168.30.0 0.0.0.255 host 192.168.30.2
 31 permit ip host 192.168.30.2 any
```

```
 35 permit ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
 40 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
 50 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
 60 permit ip any any
interface vlan 30
 ip access-group ACL_ADMINISTRATION in
```

11.4 ACL Test Results

Source	Destination	Expected	Result
HR PC (VLAN 20)	ping 192.168.30.2 (Web Server)	PASS	OK
HR PC (VLAN 20)	ping 192.168.10.3 (DNS)	PASS	OK
HR PC (VLAN 20)	ftp 192.168.10.4 (FTP)	BLOCKED	OK
HR PC (VLAN 20)	ping 192.168.30.10 (Admin PC)	BLOCKED	OK
Admin PC (VLAN 30)	ping 192.168.30.2 (Web Server)	PASS	OK
Admin PC (VLAN 30)	ftp 192.168.10.4 (FTP)	BLOCKED	OK
Admin PC (VLAN 30)	ping 192.168.20.10 (HR PC)	BLOCKED	OK
IT PC (VLAN 10)	ping 192.168.20.10 (HR PC)	PASS	OK
IT PC (VLAN 10)	ping 192.168.30.10 (Admin PC)	PASS	OK
IT PC (VLAN 10)	ftp 192.168.10.4	PASS	OK

12. Project Status

Phase	Description	Status
Core Layer	C1, C2, WAN, OSPF, NAT	Completed
Distribution Layer	R1, R2, R3, OSPF	Completed
Access Layer L3	S1, S2, S3, SVI, DHCP relay	Completed
Access Layer L2	A1-A6, VLAN segmentation	Completed
Endpoints & IP Addressing	Servers, PCs, DHCP pools	Completed
DNS	Internal A records for services	Completed
FTP/NAS	Users and permissions	Completed
Web Server (Intranet)	HTTP/HTTPS on VLAN 30	Completed
NAT	PAT on C1 and C2 toward WAN	Completed
Inter-VLAN ACLs	ACL_HR and ACL_ADMINISTRATION	Completed
End-to-End Testing	Ping, HTTP, FTP across VLANs	Completed